09-18 第1篇

阅读理解（说明文） Why Is the Sky Blue — and Why Does the Sunset Turn Red?

① Every morning millions of people open their curtains and look up at a **soft blue sky**. Every evening, if the weather is kind, that same sky slowly changes into **soft orange, warm pink, and deep red** before the night arrives. Small children often ask, "Why is the sky blue in the daytime and red at sunset?" The answer seems simple, but it tells a neat little story about **light** — and scientists needed nearly a hundred years of careful experiments to fully understand it.

② The first important fact is that **sunlight is not white**. Although it looks white to our eyes, ordinary sunlight is actually a **mixture** of many colours: violet, blue, green, yellow, orange and red. Isaac Newton, the famous English scientist, showed this in the 17th century by letting sunlight pass through a glass **prism**. The light came out on the other side separated into a small rainbow — the very same colours our eyes usually mix back together as "white".

③ The second important fact is about what happens when this mixed light enters our atmosphere. Our air is full of extremely small **gas particles**, mostly nitrogen and oxygen. When sunlight hits these particles, the different colours are **scattered** — that is, bounced in different directions — but **not equally**. Blue and violet light are scattered **much more strongly** than red and orange light, because they travel in shorter waves. This means that when you look up in the middle of the day, your eyes are actually catching blue light that has been **bounced around the sky** many times before reaching you. That is why the sky appears blue.

④ "But why not violet?" a careful reader may ask. Violet light is scattered even more than blue. Two things, however, make the sky look blue rather than violet. First, the **sun emits less violet** than blue. Second, and more importantly, **human eyes are less sensitive to violet** than to blue — our eyes are simply better at noticing blue, so that is the colour our brain "chooses" to see. Nature proposes, but our eyes choose.

⑤ At **sunset**, the geometry changes. The sun is no longer high above your head; it is close to the horizon. Sunlight must now pass through a **much thicker layer of atmosphere** before reaching your eyes — sometimes **forty times** thicker than at noon. Along that long path, the blue and violet light are scattered away so many times that almost none of them reach your eyes directly. What does reach your eyes is mostly the **orange and red** light, because these longer-wave colours can travel through the thick air with less scattering. That is why the setting sun, and the clouds near it, appear warm red and gold.

⑥ This story also explains a few extra puzzles. **Dust** and tiny water drops from forest fires or large storms make the sky redder at sunset, because more blue is scattered away. On the **Moon**, astronauts reported that the sky is black even in the daytime, because the Moon has almost no atmosphere at all to scatter the sunlight; with no scattering, there is no blue sky. And deep in the **ocean**, below a few hundred metres, the water itself absorbs red light first, and we are left in a cold blue-green world — another story of light and distance.

⑦ So the next time a small child points up and asks why the sky is blue, you can tell them a simple but surprisingly deep truth: **the sky is not really coloured at all**. Its colour is a daily gift from sunlight, from the tiny molecules of our air, and from the clever, choosing eye of every person looking up.

注释： prism 棱镜；scatter 散射；nitrogen 氮；oxygen 氧；horizon 地平线；atmosphere 大气层

根据文章内容，选择最佳答案。

1. What did Newton's prism experiment show, according to Paragraph 2?

A. That sunlight is actually dark black.
B. That ordinary "white" sunlight is really a mixture of many colours — violet, blue, green, yellow, orange and red.
C. That sunlight can be turned into electricity.
D. That only one colour of light comes from the sun.

2. According to Paragraph 3, why does the sky look blue during the day?

A. Because the sky is made of blue crystals.
B. Because gas particles in the air scatter blue light much more strongly than red and orange light, so the eye catches bounced blue light many times.
C. Because the Sun itself is blue.
D. Because blue clouds float above us.

3. Why does the sky look blue rather than violet, although violet is scattered even more?

A. Because violet is reflected back into space.
B. Because the sun emits less violet than blue, and human eyes are less sensitive to violet than to blue.
C. Because violet can only be seen by animals.
D. Because violet only appears at night.

4. According to Paragraph 5, why does the sunset look red?

A. Because the sun changes colour when it is tired.
B. At sunset, sunlight passes through a much thicker layer of atmosphere, so blue and violet are scattered away, while the longer-wave orange and red light reach our eyes.
C. Because clouds paint themselves red each evening.
D. Because red lamps are turned on near the horizon.

5. What did astronauts report about the sky on the Moon (Paragraph 6)?

A. The Moon sky is always bright green.
B. The Moon sky is black even in the daytime, because the Moon has almost no atmosphere to scatter sunlight.
C. The Moon sky is the same blue as on Earth.
D. The Moon sky is full of clouds.

答案与解析

1.
【答案】B
【题型】细节理解题
【定位】第②段 "ordinary sunlight is actually a mixture of many colours ... Isaac Newton ... prism"
【解析】棱镜实验证明阳光由多色光组成。A、C、D 均错。故选 B。

2.
【答案】B
【题型】细节理解题
【定位】第③段 "Blue and violet light are scattered much more strongly ... bounced around the sky many times"
【解析】蓝光被空气分子强散射到四处，使天空呈蓝色。A、C、D 均错。故选 B。

3.
【答案】B
【题型】细节理解题
【定位】第④段 "the sun emits less violet than blue ... human eyes are less sensitive to violet"
【解析】两大原因：阳光紫色分量少、人眼对紫光不敏感。A、C、D 均错。故选 B。

4.
【答案】B
【题型】细节理解题
【定位】第⑤段 "much thicker layer of atmosphere ... the longer-wave colours can travel through the thick air with less scattering"
【解析】日落时阳光穿透更厚的大气层，蓝紫散失、红橙色留下。A、C、D 均错。故选 B。

5.
【答案】B
【题型】细节理解题
【定位】第⑥段 "the sky is black even in the daytime, because the Moon has almost no atmosphere"
【解析】月球几乎无大气，无散射故白天也黑。A、C、D 均错。故选 B。